

Exp No: 27

BINARY SEMAPHORE IN SYSTEMC

```
WhatsApp
mithra@DESKTOP-6DUTPF3: ~
GNU nano 8.7.1 binary_semaphore.cpp
#include <systemc.h>
SC_MODULE(binary_semaphore) {
    int sem;
    sc_event ev;
    void task1() {
        while(true) {
            wait(1, SC_SEC);
            if(sem == 1) {
                sem = 0;
                cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(2, SC_SEC);
                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }
    void task2() {
        while(true) {
            wait(2, SC_SEC);
            if(sem == 1) {
                sem = 0;
                cout << "Task2 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(1, SC_SEC);
            }
        }
    }
};
```

```
WhatsApp
mithra@DESKTOP-6DUTPF3: ~
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Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
mithra@DESKTOP-6DUTPF3:~/exp25$ nano producer_consumer.cpp
mithra@DESKTOP-6DUTPF3:~/exp25$ mkdir ~/exp27
mithra@DESKTOP-6DUTPF3:~/exp25$ cd ~/exp27
mithra@DESKTOP-6DUTPF3:~/exp27$ pwd
/home/mithra/exp27
mithra@DESKTOP-6DUTPF3:~/exp27$ nano binary_semaphore.cpp
mithra@DESKTOP-6DUTPF3:~/exp27$ g++ binary_semaphore.cpp -lsystemc -o binary_semaphore
mithra@DESKTOP-6DUTPF3:~/exp27$ ls
binary_semaphore  binary_semaphore.cpp
mithra@DESKTOP-6DUTPF3:~/exp27$ ./binary_semaphore

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
mithra@DESKTOP-6DUTPF3:~/exp27$
```